

SECTION A*Answer all questions.*

1. (a) A car travels at a constant speed of 45.0 km h^{-1} around a curve in the road with a radius of 80 m.

- (i) Explain why the car is accelerating. [2]

.....

.....

.....

.....

- (ii) Calculate the angular velocity of the car (in rad s^{-1}) as it travels around the curve. [2]

.....

.....

.....

.....

- (iii) Calculate the acceleration of the car and state its direction. [3]

.....

.....

.....

.....

- (b) Discuss how the application of science enables cars to travel safely around curves. [2]

.....

.....

.....

.....